



Course Objective and Outcome Form

Department of Electrical and Computer Engineering

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North South University, Bashundhara, Dhaka-1229, Bangladesh

1. **Course Number and Title:** CSE311 - Database Management Systems
CSE311L - Database Management Systems Laboratory
2. **Number of Credits:** 3 + 0 = 3 Credits
3. **Type:** Core
4. **Prerequisites:** CSE 225
5. **Contact Hours:** 3 (Theory)+ 3 (Lab)
6. **Course Summary:** This course introduces students with database management systems for the first time in their undergraduate study. Drawbacks of flat file system are demonstrated and advantages of relational database systems are introduced. The course examines the logical organization of databases: the entity-relationship model; the hierarchical, network, and relational data models and their languages. Functional dependencies and normal forms are discussed. Design, implementation, and optimization of query languages; security and integrity; concurrency control, different level of indices, e.g., tree and hash based indices are introduced. Access costs are compared for different alternatives. This course has separate mandatory laboratory sessions every week in a separate course CSE 311L which has 0 credits, but the students (in group) use hands on SQL queries and as a culmination, they build a full-fledged database system including a front end. The evaluation of the lab works is carried over to the theory part of the course.
7. **Course Objectives:**

The objectives of this course are

 - a. to make students comprehend the advantages of using database system over flat files.
 - b. to get students familiar with requirement analysis specially data requirements of an organization
 - c. To introduce the conceptual design from requirement analysis using E-R diagrams and also mapping ER diagrams into relational schema.
 - d. to introduce the basics and usage of relational algebra that are the foundation of SQL.
 - e. to transform a relational design into physical database design using popular commercialized database, e.g., Oracle, MySQL etc.
 - f. to demonstrate and show the evils of redundancy by introducing the concepts of functional dependencies and their types.
 - g. to design full-fledged physical database systems with least redundancy and most optimized manner.
 - h. to build their independent projects emphasizing the data requirement.

8. Course Outcomes (COs):

Upon Successful completion of this course, students will be able to:

Sl.	CO Description	Weightage (%)
1	explain conceptual model through entity, relationship diagrams (E-R diagram) and extended ER diagram	30%
2	use relational algebra in simple and complex queries based on set theory including relational model	20%
3	construct code in SQL DDL (Data Definition Language) and SQL DML (Data Manipulation Language) for table creation and query processing.	25%
4	build a Web-based relational database system, using scripting languages (e.g., PHP) and an open-source database management system (e.g., MySQL).	25%

9. Mapping of CO-PO:

Sl.	CO Description	PO	KP	Bloom's taxonomy domain/ level	Delivery methods and activities	Assessment tools
CO1	explain conceptual model through entity, relationship diagrams (E-R diagram) and extended ER diagram, including relational model.	a	K3	C2	Lectures, Notes	Exam
CO2	use relational algebra in simple and complex queries based on set theory	a	K3	C3	Lectures, Notes	Exam
CO3	construct code in SQL DDL (Data Definition Language) and SQL DML (Data Manipulation Language) for table creation and query processing.	c	K5	C3	Lectures, Notes, Lab	Exam
CO4	build a Web-based relational database system, using scripting languages (e.g., PHP) and an open-source database development system (e.g., MySQL).	e	K6	C6	Lectures, Notes	Report and Demonstration

10. Resources

Text books:

No	Name of Author(s)	Year of Publication	Title of Book	Edition	Publisher's Name	ISBN
1	Abraham Silberschatz Henry F. Korth, S. Sudarshan	2019	Database System Concepts	7 th .	McGraw-Hill	ISBN-13: 978-0073523323

Reference books:

No	Name of Author(s)	Year of Publication	Title of Book	Edition	Publisher's Name	ISBN
1	Ramez Elmasri, Shamkant B. Navathe	2015	Fundamentals of Database Systems	7 th	Pearson	ISBN-13: 978-0133970777
2	Raghu Ramakrishnan Johannes Gehrke	2003	Database Management Systems	3 rd	McGraw-Hill	ISBN-13: 978-0072465631
3	Timothy Boronczyk, Elizabeth Naramore, Jason Gerner <i>et al.</i>	2009	Beginning PHP 6, Apache, MySQL 6 Web Development	1 st .	Wiley	ISBN-13: 978-0470391143

11. Weightage Distribution among Assessment Tools

Assessment Tools	Weightage (%)
Attendance and class performance	5
Quizzes	15 (Best 3 out of 4 Quizzes)
Midterm	20
Final Exam	25
Lab Work	20
Term Project	15
Total	100

12. **Grading policy:** As per NSU grading policy available in

<http://www.northsouth.edu/academic/grading-policy.html>

13. Examinations schedule

Assessment Tools	Date
Attendance	
Class Performance	Instant
Quizzes: QUIZ 1	Week 3
Quizzes: QUIZ 2	Week 5
Midterm	Week 6
Quizzes: QUIZ 3	Week 8
Quizzes: QUIZ 4	Week 11
Final Exam	As per NSU Schedule
Result	As per NSU academic calendar